

Sam Foxman

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PROFILE

I am an senior undergrad in Computer Science at Caltech. I am interested in novel applications of computer science across all domains such as aerospace, communications, cybersecurity, and AI. Following my undergrad, I hope to become a PhD and pursue a career in academia.

EDUCATION

- **Caltech** September 2022 - June 2026
B.S. in Computer Science Pasadena CA, USA

EXPERIENCE

- **Gharib Lab, Caltech** [🌐] November 2023 - Present
Undergraduate Researcher Pasadena CA, USA
 - Created GPU-accelerated image processing software to analyze fluid flow 1000x faster than any commercially available software, 2 papers & 2 patents
 - Created software for low-latency vision language models
- **Information Processing Group, NASA Jet Propulsion Lab** [🌐] June 2023 - Present
Undergraduate Researcher Pasadena CA, USA
 - Created a program to encode and decode laser signals as part of the Deep Space Optical Communications team which allows data to be downloaded 100x faster than radio
 - Program receives signals from the NASA Psyche spacecraft and from astronauts on the Orion spaceship
 - * [NASA laser message beams video of a cat named Taters back to Earth, and it's a big deal - CNN](#)
 - Designed & created GPU-based optical communications receiver
 - * Achieved world record for the highest link difficulty (bitrate · distance²) of any real-time communication system (75 Mbps @ 2.25 · 10⁸ km), outperforming previous FPGA-based systems
- **NASA Advanced Supercomputer Center** [🌐] June 2024 - Present
Research Associate Moffett Field CA, USA
 - Created an AI chatbot and search engine app that cites millions of scientific and business documents
 - [Deploying the app across NASA](#), currently beta testing at Ames Supercomputing Division
- **Caltech Infrared Processing and Analysis Center** [🌐] February 2024 - June 2025
Mission Operations and Ground Data Systems Pasadena CA, USA
 - Wrote software for the Lunar Trailblazer spacecraft, launched February 2025
 - Mission control for Lunar Trailblazer on launch day, monitoring spacecraft telemetry
 - * Rapidly wrote software to [decode anomalous telemetry the day after launch](#)
- **NVIDIA** [🌐] June 2025 - September 2025
Software Engineering Intern Santa Clara CA, USA
 - Systems software for autonomous vehicles (Mercedes-Benz & Jaguar)
 - Improved camera and radar processing verification
- **Nelson Lab, Caltech** [🌐] September 2024 - Present
Undergraduate Researcher Pasadena CA, USA
 - Designed software system for high-throughput electron diffraction data processing
- **Feinberg Lab, Yale** [🌐] May 2020 - June 2023
High School Intern New Haven CT, USA
 - Created data analysis pipelines for the Fineberg Laboratory at Yale University for Bayesian analysis of the Social Valuation Task, neuroeconomic task to measure suicide ideation
 - Presented findings in a short talk/poster at national Technology in Psychiatry (TIPS) conference

LEADERSHIP EXPERIENCE

- **Teaching Assistant, Caltech** September 2023 - June 2025
Pasadena CA, USA
Head Teaching Assistant
 - CS 171: *Computer Graphics Laboratory* TA Fall 2024, Head TA Fall 2025, 2026
 - CS 179: *GPU Programming* TA Fall 2024, Head TA Spring 2025, 2026
 - * Created assignment where students program a large language model from scratch in CUDA
- **Caltech Undergraduate Computer Science Club [🌐]** September 2023 - Present
Pasadena CA, USA
President
 - Managed a club budget of $\approx$ \$24,000, organizing club events, and maintaining relations with external sponsors, including NVIDIA and Relativity Space, interested in giving and working with our club.
 - For instance, we worked on purchasing computer parts and getting sponsored GPUs from NVIDIA, built a computer cluster from these parts, and now actively operate it for club members.
- **Caltech Aerospace Club [🌐]** September 2023 - Present
Pasadena CA, USA
Co-chair
 - Won \$150k NASA grant to study an inflatable Lunar dust shield. I lead the fluids team running computational fluid dynamics simulations. Presented at AIAA Ascend.

PUBLICATIONS

- S. Boltt, S. Foxman, M. Gharib. "RapidPIV: Full Flow-Field kHz PIV for Real-Time Display and Control". Submitted to *Measurement Science & Technology*, 2025. Preprint <https://www.arxiv.org/abs/2504.17987>.
- L. Coffin, A.J. Torres, S. Wallen, P. Calub, K. Gauld, I. Kwaterski, S. Foxman, H. Ramsperger, E. Xu, ..., K. Carpenter, S.J. Chung. "PILLARS: Plume-Deployed Inflatable for Launch and Landing Abrasive Regolith Shielding". *AIAA Aviation Forum and Ascend*, July 2025. <https://doi.org/10.2514/6.2025-4126>
- B. Ehlmann, ..., S. Foxman, ..., and the Lunar Trailblazer Team. "Lunar Trailblazer Spacecraft Tracking and Mission Recovery Attempt: Characterization of Status and Behavior of a Non-Cooperative Object in Cis-Lunar Space". Manuscript submitted, August 2025
- D. Eremin, K. Jha, D. Delgadillo, H. Zhang, S. Foxman, S. Johnson, N. Vlahakis, D. Cascio, V. Lavallo, J. Rodriguez, H. Nelson. "Spatially-Aware Diffraction Mapping Enables Fully Autonomous MicroED". In review at *Journal of the American Chemical Society (JACS)*, 2025. Preprint <https://doi.org/10.26434/chemrxiv-2025-4p4c3>
- B. Ehlmann, ..., S. Foxman, ..., and the Lunar Trailblazer Team. "The Lunar Trailblazer Mission: Science Motivation and Implementation of a Pioneering Small Satellite for Lunar Water and Lunar Geology in the NASA SIMPLEX program". Submitted to *Journal of Geophysical Research*, August 2025
- S. Foxman, D. Dalle. "NASA-GPT". Manuscript in preparation. September 2025
- S. Foxman, R. Rogalin. "GPU Receivers for Deep Space Optical Communications". Presenting at SPIE Photonics West, Free-Space Laser Communications XXXVIII, January 2026.

INVENTIONS

- S. Boltt, S. Foxman, M. Gharib. "Portable Real-Time Optical Flow-Field Sensor". U.S. Patent Application No. 63/715,303. November 2024
- S. Foxman, R. Rogalin. "GPU-Based Optical Communications Modem". Software Copyright Caltech/NASA Jet Propulsion Laboratory. June 2025
- S. Foxman, S. Boltt, M. Gharib. "Systems and Methods for Optical Flow Using GPU Tensor Processing Cores". U.S. Patent Application No. 63/823,578. June 2025

SKILLS

- **Programming Languages and Frameworks:** C++, Linux, Java, Python, Rust, CUDA, OpenGL, MATLAB, Jupyter, C, JavaScript, Full-Stack Development, Computer Networking, R, SQL, Bash
- **General:** Sysadmin, Cloud Servers, CAD, Embedded Systems, FPGA Software Programming, Robotics, 3D Printing, EEG Preprocessing Pipeline, HPC (High performance computing), Information Retrieval, AI, Wireless Communication